STAT/MA 41600
In-Class Problem Set #24: October 14, 2026
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Problem Set 24 Answers

1a. We solve for a by computing $0.75 = P(X > a) = \int_a^\infty \frac{1}{10} e^{-x/10}\, dx = -e^{-x/10}|_{x=a}^\infty = e^{-a/10}$. So we conclude that $a = -10 \ln(0.75) = 2.8768$.

1b. We have $F_X(a) = 0$ for $a \leq 0$. For $a > 0$, we have $F_X(a) = P(X \leq a) = \int_0^a \frac{1}{10} e^{-x/10}\, dx = -e^{-x/10}|_{x=0}^a = 1 - e^{-a/10}$. So we conclude that $F_X(x) = 1 - e^{-x/10}$ for $x > 0$, and $F_X(x) = 0$ otherwise.

2a. We need $1 = \int_{-1}^3 f_Y(y)\, dy = \int_{-1}^3 (k)(y+1)(y-3)^2 dy = (k)(\frac{1}{4}y^4 - \frac{5}{3}y^3 + \frac{3}{2}y^2 + 9y)|_{y=-1}^3 = (k)(64/3)$, so $k = 3/64 = 0.046875$.

2b. Yes, we have such a k. We have $1 = \int_{-1}^3 (k)(y+1)(y-3)\, dy = (k)(\frac{1}{3}y^3 - y^2 - 3y)|_{y=-1}^3 = (k)(-32/3)$, so $k = -3/32 = -0.09375$.

3a. Because we have $f_X(x) = 1/7$ for $0 < x < 7$, we have $P(|X - 5| < 1/3) = (2/3)/7 = 2/21 = 0.09524$.

3b. Similarly, if $f_X(x) = 1/k$ for $0 < x < k$ (and if we assume k is at least $5\frac{1}{3}$, e.g., we are safe if $k > 7$, for instance), we have $P(|X - 5| < 1/3) = (2/3)/k = 2/(3k)$.

4. Suppose that the cumulative distribution function of X is $F_X(x) = 1 - e^{-5x/2}$ for $x > 0$, and $F_X(x) = 0$ otherwise.

4a. The pdf of X is $\frac{5}{2} e^{-5x/2}$ for $x > 0$, and $f_X(x) = 0$ otherwise.

4b. We have $P(X > a) = \int_a^\infty \frac{5}{2} e^{-5x/2}\, dx = -e^{-5x/2}|_{x=a}^\infty = e^{-5a/2}$.

5a. The median value a has $1/2 = 1 - e^{-a/10}$, so $a = -10 \ln(1/2) = 6.9315$.

5b. The median value a has $1/2 = 1 - e^{-5a/2}$, so $a = -\frac{2}{5} \ln(1/2) = 0.2773$.